sys.tables (Transact-SQL)

Applies to: SQL Server · Azure SQL Database · Azure SQL Managed Instance · Azure Synapse Analytics · Analytics Platform System (PDW) · SQL analytics endpoint in Microsoft Fabric · Warehouse in Microsoft Fabric · SQL database in Microsoft Fabric

Returns a row for each user table in a SQL Server database.

Expand table

Column name	Data type	Description
`<inherited columns>`		For a list of columns that this view inherits, see sys.objects (Transact-SQL).
`lob_data_space_id`	**int**	A nonzero value is the ID of the data space (filegroup or partition scheme) that holds the large object binary (LOB) data for this table. Examples of LOB data types include **varbinary(max)**, **varchar(max)**, **geography**, or **xml**. 0 = The table doesn't have LOB data.
`filestream_data_space_id`	**int**	The data space ID for a FILESTREAM filegroup or a partition scheme that consists of FILESTREAM filegroups. To report the name of a FILESTREAM filegroup, execute the query `SELECT FILEGROUP_NAME (filestream_data_space_id) FROM sys.tables`. `sys.tables` can be joined to the following views on `filestream_data_space_id` = `data_space_id`. - `sys.filegroups` - `sys.partition_schemes` - `sys.indexes` - `sys.allocation_units` - `sys.fulltext_catalogs` - `sys.data_spaces` - `sys.destination_data_spaces` - `sys.master_files` - `sys.database_files` - `backupfilegroup` (join on `filegroup_id`)
`max_column_id_used`	**int**	Maximum column ID ever used by this table.

Column name	Data type	Description
`lock_on_bulk_load`	**bit**	Table is locked on bulk load. For more information, see sp_tableoption (Transact-SQL).
`uses_ansi_nulls`	**bit**	Table was created with the `SET ANSI_NULLS` database option `ON`.
`is_replicated`	**bit**	1 = Table is published using snapshot replication or transactional replication.
`has_replication_filter`	**bit**	1 = Table has a replication filter.
`is_merge_published`	**bit**	1 = Table is published using merge replication.
`is_sync_tran_subscribed`	**bit**	1 = Table is subscribed using an immediate updating subscription.
`has_unchecked_assembly_data`	**bit**	1 = Table contains persisted data that depends on an assembly whose definition changed during the last `ALTER ASSEMBLY`. Will be reset to 0 after the next successful `DBCC CHECKDB` or `DBCC CHECKTABLE`.
`text_in_row_limit`	**int**	The maximum bytes allowed for text in row. 0 = Text in row option isn't set. For more information, see sp_tableoption (Transact-SQL).
`large_value_types_out_of_row`	**bit**	1 = Large value types are stored out-of-row. For more information, see sp_tableoption (Transact-SQL).
`is_tracked_by_cdc`	**bit**	1 = Table is enabled for change data capture. For more information, see sys.sp_cdc_enable_table (Transact-SQL).
`lock_escalation`	**tinyint**	A value of the `LOCK_ESCALATION` option for the table: 0 = `TABLE` 1 = `DISABLE` 2 = `AUTO`
`lock_escalation_desc`	**nvarchar(60)**	A text description of the lock_escalation option for the table. Possible values are: `TABLE`, `AUTO`, and `DISABLE`.
`is_filetable`	**bit**	1 = Table is a FileTable. For more information about FileTables, see FileTables (SQL Server).

Column name	Data type	Description
		Applies to: SQL Server 2012 (11.x) and later versions, and Azure SQL Database
`is_memory_optimized`	**bit**	The following are the possible values: 0 = not memory optimized. 1 = is memory optimized. A value of 0 is the default value. Memory optimized tables are in-memory user tables, the schema of which is persisted on disk similar to other user tables. Memory optimized tables can be accessed from natively compiled stored procedures. **Applies to**: SQL Server 2014 (12.x) and later versions, and Azure SQL Database.
`durability`	**tinyint**	The following are possible values: 0 = `SCHEMA_AND_DATA` 1 = `SCHEMA_ONLY` A value of `0` is the default value. **Applies to**: SQL Server 2014 (12.x) and later versions, and Azure SQL Database

`durability_desc`	**nvarchar(60)**	The following are the possible values:
		
		`SCHEMA_ONLY`
		`SCHEMA_AND_DATA`
		
		A value of `SCHEMA_AND_DATA` indicates that the table is a durable, in-memory table. `SCHEMA_AND_DATA` is the default value for memory optimized tables. A value of `SCHEMA_ONLY` indicates that the table data isn't persisted upon restart of the database with memory optimized objects.
		
		Applies to: SQL Server 2014 (12.x) and later versions, and Azure SQL Database

`temporal_type`	**tinyint**	The numeric value representing the type of table: 0 = `NON_TEMPORAL_TABLE` 1 = `HISTORY_TABLE` (associated with a temporal table) 2 = `SYSTEM_VERSIONED_TEMPORAL_TABLE`

Column name	Data type	Description
		Applies to: SQL Server 2016 (13.x) and later versions, and Azure SQL Database

`temporal_type_desc`	**nvarchar(60)**	The text description of the type of table:
		
		`NON_TEMPORAL_TABLE`
		`HISTORY_TABLE`
		`SYSTEM_VERSIONED_TEMPORAL_TABLE`
		
		Applies to: SQL Server 2016 (13.x) and later versions, and Azure SQL Database

`history_table_id`	**int**	When `temporal_type` is `2` or `ledger_type` is `2`, returns `object_id` of the table that maintains historical data for a temporal table, otherwise returns `NULL`. **Applies to**: SQL Server 2016 (13.x) and later versions, and Azure SQL Database
`is_remote_data_archive_enabled`	**bit**	Indicates whether the table is Stretch-enabled. 0 = The table isn't Stretch-enabled. 1 = The table is Stretch-enabled. For more info, see Stretch Database. **Applies to**: SQL Server 2016 (13.x) and later versions, and Azure SQL Database
`is_external`	**bit**	Indicates table is an external table. 0 = The table isn't an external table. 1 = The table is an external table. **Applies to**: SQL Server 2016 (13.x) and later versions, Azure SQL Database, and Azure Synapse Analytics
`history_retention_period`	**int**	The numeric value representing duration of the temporal history retention period in units specified with `history_retention_period_unit`. **Applies to**: SQL Server 2017 (14.x) and later versions, and Azure SQL Database
`history_retention_period_unit`	**int**	The numeric value representing type of temporal history retention period unit.

Column name	Data type	Description
		-1: `INFINITE` 0: `SECOND` 1: `MINUTE` 2: `HOUR` 3: `DAY` 4: `WEEK` 5: `MONTH` 6: `YEAR` **Applies to**: SQL Server 2017 (14.x) and later versions, and Azure SQL Database

`history_retention_period_unit_desc`	**nvarchar(10)**	The text description of type of temporal history retention period unit.
		
		`INFINITE`
		`SECOND`
		`MINUTE`
		`HOUR`
		`DAY`
		`WEEK`
		`MONTH`
		`YEAR`
		
		Applies to: SQL Server 2017 (14.x) and later versions, and Azure SQL Database

`is_node`	**bit**	1 = Graph node table. 0 = Not a graph node table. **Applies to**: SQL Server 2017 (14.x) and later versions, and Azure SQL Database
`is_edge`	**bit**	1 = Graph edge table. 0 = Not a graph edge table. **Applies to**: SQL Server 2017 (14.x) and later versions, and Azure SQL Database
`data_retention_period`	**int**	The numeric value representing duration of the data retention period in units specified with `data_retention_period_unit`. **Applies to**: Azure SQL Edge only
`data_retention_period_unit`	**int**	The numeric value representing type of data retention period unit.

Column name	Data type	Description
		-1: `INFINITE` 0: `SECOND` 1: `MINUTE` 2: `HOUR` 3: `DAY` 4: `WEEK` 5: `MONTH` 6: `YEAR` **Applies to**: Azure SQL Edge only

`data_retention_period_unit_desc`	**nvarchar(10)**	The text description of type of data retention period unit.
		
		`INFINITE`
		`SECOND`
		`MINUTE`
		`HOUR`
		`DAY`
		`WEEK`
		`MONTH`
		`YEAR`
		
		Applies to: Azure SQL Edge only

`ledger_type`	**tinyint**	The numeric value indicates if the table is a ledger table. 0 = `NON_LEDGER_TABLE` 1 = `HISTORY_TABLE` (associated with an updatable ledger table) 2 = `UPDATABLE_LEDGER_TABLE` 3 = `APPEND_ONLY_LEDGER_TABLE` For more information on database ledger, see Ledger. **Applies to**: SQL Server 2022 (16.x) and later versions, and Azure SQL Database

`ledger_type_desc`	**nvarchar(60)**	The text description of a value in the `ledger_type` column:
		
		`NON_LEDGER_TABLE`
		`HISTORY_TABLE`
		`UPDATABLE_LEDGER_TABLE`

Column name	Data type	Description
		`APPEND_ONLY_LEDGER_TABLE` **Applies to**: SQL Server 2022 (16.x) and later versions, and Azure SQL Database
`ledger_view_id`	**int**	When `ledger_type IN (2, 3)` returns `object_id` of the ledger view, otherwise returns `NULL`. **Applies to**: SQL Server 2022 (16.x) and later versions, and Azure SQL Database
`is_dropped_ledger_table`	**bit**	Indicates a ledger table that was dropped. **Applies to**: SQL Server 2022 (16.x) and later versions, and Azure SQL Database

Permissions

The visibility of the metadata in catalog views is limited to securables that a user either owns, or on which the user was granted some permission. For more information, see Metadata Visibility Configuration.

Examples

A. Return all user tables without a primary key

The following example returns all of the user tables that don't have a primary key.

SQL

```
SELECT SCHEMA_NAME(schema_id) AS schema_name, name AS table_name
FROM sys.tables
WHERE OBJECTPROPERTY(object_id, 'TableHasPrimaryKey') = 0
ORDER BY schema_name, table_name;
GO
```

B. List temporal data related tables

The following example shows how related temporal data can be exposed.

Applies to: SQL Server 2016 (13.x) and later versions, and Azure SQL Database.

SQL

```
SELECT T1.object_id,
    T1.name AS TemporalTableName,
    SCHEMA_NAME(T1.schema_id) AS TemporalTableSchema,
    T2.name AS HistoryTableName,
    SCHEMA_NAME(T2.schema_id) AS HistoryTableSchema,
    T1.temporal_type_desc
FROM sys.tables T1
LEFT JOIN sys.tables T2
    ON T1.history_table_id = T2.object_id
ORDER BY T1.temporal_type DESC;
```

C. List information about temporal history retention

The following example shows how information on temporal history retention can be exposed.

Applies to: SQL Server 2017 (14.x) and later versions, and Azure SQL Database.

SQL

```
SELECT DB.is_temporal_history_retention_enabled,
    SCHEMA_NAME(T1.schema_id) AS TemporalTableSchema,
    T1.name AS TemporalTableName,
    SCHEMA_NAME(T2.schema_id) AS HistoryTableSchema,
    T2.name AS HistoryTableName,
    T1.history_retention_period,
    T1.history_retention_period_unit_desc
FROM sys.tables T1
OUTER APPLY (
    SELECT is_temporal_history_retention_enabled
    FROM sys.databases
    WHERE name = DB_NAME()
) DB
LEFT JOIN sys.tables T2
    ON T1.history_table_id = T2.object_id
WHERE T1.temporal_type = 2;
```

Related content

- Object Catalog Views (Transact-SQL)
- System catalog views (Transact-SQL)
- DBCC CHECKDB (Transact-SQL)
- DBCC CHECKTABLE (Transact-SQL)
- Querying the SQL Server System Catalog FAQ
- In-Memory OLTP overview and usage scenarios

Last updated on 11/18/2025